Hall-Littlewood Symmetric Functions

0. Introduction.

Let $x_1, \dots, x_n, t$ be commuting variables over $\mathbb{Z}$. The Hall-Littlewood polynomials $P_\lambda(x_1, \dots x_n, t)$ are symmetric functions which interpolate between the Schur polynomials s_λ and the monomial symmetric function m_λ:

$$P_\lambda(x_1, \dots x_n, 0) = s_\lambda(x_1, \dots x_n)$$

$$P_\lambda(x_1, \dots, x_n, 1) = m_\lambda(x_1, \dots, x_n).$$

Moreover, they form a $\mathbb{Z}[t]$-basis of $\Lambda[t]$ and we have:

$$s_\lambda(x_1, \dots, x_n) = \sum_\mu K_{\lambda\mu}(t)\, P_\mu(x_1, \dots x_n, t),$$

so that the above is a t-analogue of the classical expression:

$$s_\lambda(x) = \sum_\mu K_{\lambda\mu}\, m_\mu,$$

where $K_{\lambda\mu}(t)$ are the so called Kostka-Foulkes polynomials.

1. The polynomials R_λ.

Let λ be a partition with $\ell(\lambda) \leq n$. Define:

$$R_\lambda(x_1, \dots, x_n; t) = \sum_{w \in S_n} w \cdot \left(x_1^{\lambda_1} \cdots x_n^{\lambda_n} \cdot \prod_{i<j} \frac{x_i - t x_j}{x_i - x_j} \right).$$

Ex. $R_{\square\square}(x_1, x_2; t) = x_1^2 + x_1 x_2 + x_2^2 - t x_1 x_2$.

Since $a_\delta = \prod_{i<j} (x_i - x_j)$, we rewrite

$$R_\lambda(x_1, \dots, x_n; t) = \frac{1}{a_\delta} \sum_{w \in S_n} \operatorname{sgn}(w)\, w \cdot \left(x^\lambda \prod_{i<j} (x_i - t x_j) \right) \quad (*)$$

Prop. i) $R_\lambda(x; t)$ is a hom. symmetric polynomial in $x_1, \dots, x_n$ of degree $|\lambda|$ with coeff. in $\mathbb{Z}[t]$.

ii) $R_\lambda(x; t) = \sum_\mu u_{\lambda\mu}(t)\, s_\mu(x)$, where

$u_{\lambda\mu}(t) = 0$ unless $|\lambda| = |\mu|$ and $\mu \trianglelefteq \lambda$. Moreover,

$$u_{\lambda\lambda}(t) = \prod_{i \geq 0} [m_i]_t!$$

with $m_i = \#$ parts of λ equal to i, $[n]_t = 1 + t + t^2 + \cdots + t^{n-1}$.

iii) $R_\lambda(x; 0) = s_\lambda(x)$, $R_\lambda(x; 1) = |S_n^\lambda|\, m_\lambda(x)$.

Proof.

i) The sum in the RHS of (*) is anti-symmetric in x_i's, whence it is divisible by a_δ in $\mathbb{Z}[x_1, \dots, x_n, t]$. Thus, the RHS of (*) is a hom. symmetric polynomial with coeff. in $\mathbb{Z}[t]$.

iii) We have $R_\lambda(x,0) = \frac{1}{a_\delta} \sum_{w \in S_n} \operatorname{sgn}(w)\, w \cdot x^{\lambda+\delta} = \frac{a_{\lambda+\delta}}{a_\delta} = s_\lambda$.

Also, $R_\lambda(x,1) = \sum_{w \in S_n} w \cdot x^\lambda = |S_n^\lambda|\, m_\lambda$.

ii) Any of the summands in $\prod_{i<j} (x_i - t x_j)$ is of the form:

$$\prod_{i<j} x_i^{r_{ij}} (-t x_j)^{r_{ji}}$$

for some $n \times n$ matrix (r_{ij}) of 0's and 1's such that:

$$r_{ii} = 0, \quad r_{ij} + r_{ji} = 1, \quad i \neq j.$$

Hence, we have:

$$x^\lambda \cdot \prod_{i<j} x_i^{r_{ij}} (-t x_j)^{r_{ji}} = (-t)^d x^\alpha$$

where $\alpha = \lambda_i + \sum_j r_{ij}$ and $d = \sum_{i<j} r_{ji}$.

Hence, R_λ is a sum of terms of the form:

$$(-t)^d a_\alpha \cdot a_\delta^{-1}.$$

Since $a_\alpha = 0$ if α has repetitions, we may assume that

$$\alpha_{w(1)} > \alpha_{w(2)} > \cdots > \alpha_{w(n)} \geq 0 \quad \text{for some } w \in S_n.$$

Thus, $w \cdot \alpha = \mu + \delta$ for some partition μ, whence

$$a_\alpha \cdot a_\delta^{-1} = \operatorname{sgn}(w)\, s_\mu.$$

We now show that $\mu \trianglelefteq \lambda$. Set (s_{ij}) to be the $n \times n$ matrix

$$s_{ij} = r_{w(i), w(j)}$$

so that $s_{ii} = 0$, $s_{ij} + s_{ji} = 1$ $(i \neq j)$, and

$$\mu_i + n - i = \alpha_{w(i)} = \lambda_{w(i)} + \sum_j s_{ij}.$$

Thus, $\mu_1 + \cdots + \mu_k = \lambda_{w(1)} + \cdots + \lambda_{w(k)} + \sum_{i=1}^{k} \sum_{j=1}^{n} s_{ij} - \sum_{i=1}^{k} (n-i)$.

In particular, $|\mu| = |\lambda|$.

Now, we have:

$$\sum_{i=1}^{k} \sum_{j=1}^{n} s_{ij} = \underbrace{\sum_{i=1}^{k} \sum_{j=1}^{k} s_{ij}}_{= \frac{1}{2} k(k-1)} + \underbrace{\sum_{i=1}^{k} \sum_{j=k+1}^{n} s_{ij}}_{\leq k(n-k), \text{ as } s_{ij} \leq 1}$$

$$\leq \frac{1}{2} k(k-1) + k(n-k) = \sum_{i=1}^{k} (n-i).$$

Hence, $\mu_1 + \cdots + \mu_k \leq \lambda_{w(1)} + \cdots + \lambda_{w(k)} \leq \lambda_1 + \cdots + \lambda_k$, so $\mu \trianglelefteq \lambda$.

Thus, $R_\lambda(x;t)$ is a sum of elements of the form $(-t)^d \operatorname{sgn}(w)\, s_\mu$, $\mu \trianglelefteq \lambda$, $|\mu| = |\lambda|$.

We now compute the coefficient $u_{\lambda\lambda}(t)$. Indeed, we have $\lambda = \mu$ if l if $\lambda_{w(i)} = x_i$ and $s_{ij} = 1$ for all $i<j$. Hence,

$$u_{\lambda\lambda}(t) = \sum_{w \in S_n^\lambda} \operatorname{sgn}(w)\,(-t)^d,$$

with $$d = \sum_{i<j} r_{ji} = \sum_{i<j} s_{w^{-1}(j),w^{-1}(i)} = n(w^{-1}),$$

where $n(w^{-1}) = \#\{i<j : w^{-1}(j) < w^{-1}(i)\}$ is the number of inversions of w^{-1}.

Since $n(w) = n(w^{-1})$ and $\operatorname{sgn}(w) = (-1)^{n(w)}$, then

$$u_{\lambda\lambda}(t) = \sum_{w \in S_n^\lambda} t^{n(w)}.$$

<u>Lemma</u> $$\sum_{w \in S_m} t^{n(w)} = 1(1+t)\cdots(1+t+\cdots+t^{m-1}) = [m]_t!$$

Construct a bijection $I: \{0\}\times\{0,1\}\times\cdots\times\{0,\ldots,m-1\} \to S_m$ as follows. For $(s_1,\ldots,s_m) \in \{0\}\times\{0,1\}\times\cdots\times\{0,\ldots,m-1\}$, we set $I(s_1,\ldots s_m)$ to be the word in $1,\ldots m$ such that i is put in the exact position such that it precedes exactly s_i elements among $1,2,\ldots,i-1$. In particular, the placement of i creates s_i new inversions and we have $n(I(s_1,\ldots s_m)) = s_1+\cdots+s_m$,

Example: $m=5$.

$$I(0,0,2,1,2) = 31542$$

Thus, $$\sum_{w \in S_m} t^{n(w)} = \sum_{(S_{i_1} \to S_m)} t^{n(I(S_{i_1} \to S_m))} = \sum_{(S_{i_1} \to S_m)} t^{S_{i_1} + \cdots + S_m} = [m]_t!$$

Since $S_n^\lambda = \prod_{i \geq 0} S_{m_i}$, then

$$u_{\lambda\lambda}(t) = \sum_{w \in S_n^\lambda} t^{n(w)} = \sum_{(w_0, w_1, \ldots)} t^{n(w_0) + n(w_1) + \cdots}$$

$$= \prod_{i \geq 0} \sum_{w_i \in S_{m_i}} t^{n(w_i)} = \prod_{i \geq 0} [m_i]_t!$$

For convenience, set $v_\lambda(t) = \prod_{i \geq 0} [m_i]_t!$

Corollary: We have $\sum_{w \in S_n} w \cdot \left(\prod_{i<j} \frac{x_i - t x_j}{x_i - x_j} \right) = [n]_t$.

Proof. Take $\lambda = 0$ in the original def of R_λ. Then, the expansion of R_λ in terms of Schur functions has only one summand equal to $[n]_t$, as $\lambda = 0$ has n parts equal to zero.

Lemma: The polynomial $R_\lambda(x_1,\dots,x_n;t)$ is divisible by $v_\lambda(t)$ in $\mathbb{Z}[t]$.

Proof. Suppose that $\lambda_{i_1} = \cdots = \lambda_{i_m}$ and let $I = \{i_1,\dots,i_m\}$. Then, any $w \in S_I$ fixes $x_1^{\lambda_1}\cdots x_n^{\lambda_n}$, and by the corollary we get:

$$R_\lambda(x_1,\dots,x_n;t) = \sum_{w\in S_n/S_I}\sum_{w'\in S_I} ww'\cdot\left(x_1^{\lambda_1}\cdots x_n^{\lambda_n}\prod_{i<j}\frac{x_i-tx_j}{x_i-x_j}\right)$$

$$= [m]_t\cdot\sum_{w\in S_n/S_I} w\cdot\left(x_1^{\lambda_1}\cdots x_n^{\lambda_n}\prod_{\substack{i<j\\(i,j)\notin I\times I}}\frac{x_i-tx_j}{x_i-x_j}\right).$$

Proceeding in this way with the remaining parts of λ, we get:

$$R_\lambda(x_1,\dots,x_n;t) = v_\lambda(t)\sum_{w\in S_n/S_n^\lambda} w\cdot\left(x_1^{\lambda_1}\cdots x_n^{\lambda_n}\prod_{\lambda_i>\lambda_j}\frac{x_i-tx_j}{x_i-x_j}\right).$$

∎

Def We define the Hall-Littlewood polynomial $P_\lambda(x_1,\dots,x_n;t)$ as the quotient:

$$P_\lambda(x_1,\dots,x_n;t) = \frac{1}{v_\lambda(t)}R_\lambda(x_1,\dots,x_n;t)$$

$$= \frac{1}{v_\lambda(t)}\sum_{w\in S_n} w\cdot\left(x^\lambda\prod_{i<j}\frac{x_i-tx_j}{x_i-x_j}\right)$$

$$= \sum_{w\in S_n/S_n^\lambda} w\cdot\left(x^\lambda\prod_{\lambda_i>\lambda_j}\frac{x_i-tx_j}{x_i-x_j}\right).$$

Thus, $P_\lambda(x;t)$ is a symm. polynomial in $x_1,\dots,x_n$ with coefficients in $\mathbb{Z}[t]$.

Prop: Let $\ell(\lambda) \leq n$. Then, $P_\lambda(x_1, \dots x_n, 0, t) = P_\lambda(x_1, \dots, x_n, t)$.

Proof

We have $P_\lambda(x_1, \dots x_{n+1}, t) = \sum_{w \in S_{n+1}/S_{n+1}^\lambda} w \cdot \left(x_1^{\lambda_1} \cdots x_{n+1}^{\lambda_{n+1}} \prod_{i<j} \frac{x_i - t x_j}{x_i - x_j} \right)$.

Setting $x_{n+1} = 0$, the only summands for which the RHS are non-zero are those for which $w \in S_{n+1}$ permutes $n+1$ with r such that $\lambda_r = 0$, which is a subset of the permutations preserving the zero parts of λ. Taking w modulo S_{n+1}^λ, then w fixes $n+1$, ∎

We define the symmetric function $P_\lambda(x, t)$ in inf. many variables as the element of $\Lambda[t]$ whose image in $\Lambda_n[t]$ is $P_\lambda(x_1, \dots x_n, t)$ for every $n \geq \ell(\lambda)$.

The following properties of $P_\lambda(x, t)$ are immediate

- $P_\lambda(x_1, \dots, x_n, 0) = s_\lambda(x_1, \dots x_n)$
- $P_\lambda(x_1, \dots, x_n, 1) = m_\lambda(x_1, \dots, x_n)$
- $P_\lambda(x_1, \dots x_n, t) = \sum_{\mu \leq \lambda} w_{\lambda\mu}(t)\, s_\mu(x_1, \dots, x_n)$,

where $w_{\lambda\mu}(t) = 0$ unless $|\lambda| = |\mu|$ and $\mu \leq \lambda$, with $w_{1\lambda}(t) = 1$.

- $P_{(1^r)}(x_1, \dots, x_n, t) = e_r(x_1, \dots x_n)$

In particular, the transition matrix which expresses P_λ in terms of the s_μ is strictly upper triangular. Since the s_μ form a $\mathbb{Z}[t]$-basis of $\Lambda[t]$, we have:

- The symm. functions $P_\lambda(x;t)$ form a $\mathbb{Z}[t]$-basis of $\Lambda[t]$.

Hence one may write the Schur function s_λ in terms of the $P_\mu(x,t)$ as follows:

$$s_\lambda(x_1,\dots,x_n) = \sum_{\mu \trianglelefteq \lambda} K_{\lambda\mu}(t)\, P_\mu(x_1,\dots,x_n,t),$$

where $K_{\lambda\mu}(t)$ are t-analogues of the Kostka numbers and $K_{\lambda\mu}(t)$ are known as the Kostka-Foulkes polynomials.

2. Hall algebras

Let G be an abelian p-group. Then,

$$G \simeq \mathbb{Z}/p^{\lambda_1} \oplus \cdots \oplus \mathbb{Z}/p^{\lambda_n},$$

where $\lambda = (\lambda_1, \ldots, \lambda_n)$ is some partition. We say G is of <u>type λ</u>.

We may view G as a $\mathbb{Z}$-module. In particular, $n \cdot G$ means the submodule generated by all elements of the form $n \cdot x$.

<u>Lemma</u> Let G be of type λ. Then,

$$G \supseteq pG \supseteq p^2 G \supseteq \cdots \supseteq p^{|\lambda|} G = \{0\}$$

with $p^{i-1}G/p^iG$ being of type $(1^{\lambda_i'})$, where $\lambda' = (\lambda_1', \ldots \lambda_s')$ is the transpose.

<u>Proof</u> Let $\langle x_j \rangle = \mathbb{Z}/p^{\lambda_j}\mathbb{Z}$. Then, $\lambda_j \geq i$ if/if $p^{i-1}\langle x_j \rangle$ is nontrivial if/if $p^i \langle x_j \rangle \leq p^{i-1}\langle x_j \rangle$ is proper if/if $p^{i-1}\langle x_j \rangle / p^i \langle x_j \rangle \simeq \mathbb{Z}/p\mathbb{Z}$.

Thus, $$p^{i-1}G/p^iG \simeq \bigoplus_{\lambda_j \geq i} \mathbb{Z}/p\mathbb{Z}$$

Since $\lambda_i' = \{j : \lambda_j \geq i\}$, the result follows.

∎

For partitions $\mu^{(1)}, \cdots, \mu^{(r)}, \lambda$, let

$$G^{\lambda}_{\mu^{(1)}, \cdots, \mu^{(r)}}$$

be the number of chain of subgroups of a type λ group G:

$$G = G_0 \underbrace{\supseteq G_1 \supseteq \cdots}_{\mu^{(1)}} \underbrace{\supseteq G_r}_{\mu^{(r)}} = \{0\}$$

such that G_{i-1}/G_i is of type $\mu^{(i)}$.

In particular, $G^{\lambda}_{\mu\nu}$ is the number of subgroups N of type ν with G/N of type μ (cotype):

$$G \underbrace{\supseteq N}_{\mu} \overbrace{\supseteq \{0\}}^{\nu}$$

Let H be a free $\mathbb{Z}$-module on a basis (u_λ) indexed by all partitions λ. Define a product on H as follows:

$$u_\mu \cdot u_\nu = \sum_\lambda G^{\lambda}_{\mu\nu} u_\lambda$$

Since $G^{\lambda}_{\mu\nu} = 0$ unless $|\lambda| = |\mu| + |\nu|$, the above sum is finite.

Prop H is a commutative and associative ring with identity, where the identity is u_0 (zero partition).

The ring H is called the (classical) Hall algebra.

Prop The Hall algebra H is generated as a $\mathbb{Z}$-algebra by the elements $u_{(1^r)}$ and they are algebraically independent over $\mathbb{Z}$.

Proof. Let λ be a partition with conjugate $\lambda' = (\lambda_1', \dots, \lambda_s')$. Consider the product over the columns of λ:

$$\pi_\lambda = u_{(1^{\lambda_1'})} \cdot u_{(1^{\lambda_2'})} \cdots u_{(1^{\lambda_s'})}.$$

Then, π_λ can be written as a linear comb. of the u_μ:

$$\pi_\lambda = \sum_\mu a_{\lambda\mu} u_\mu,$$

where $a_{\lambda\mu}$ is the number of filtrations of a type μ group

$$G = G_0 \supseteq G_1 \supseteq \cdots \supseteq G_s = \{0\}$$

such that $G_{i-1}/G_i \cong \mathbb{Z}/p\mathbb{Z}^{\oplus \lambda_i'}$ is of type $(1^{\lambda_i'})$.

If $a_{\lambda\mu} \neq 0$, then $p \cdot G_{i-1} \subseteq G_i \ \forall i$, whence $p^i G \subseteq G_i$. Thus,

$$|G/G_i| \leq |G/p^i G|.$$

Since $G/G_1 \simeq G/G_2 \Big/ G_1/G_2$, then

$$|G/G_2| = |G/G_1| \cdot |G_1/G_2| = p^{\lambda_1' + \lambda_2'}.$$

By induction, we find that $|G/G_i| = p^{\lambda_1' + \cdots + \lambda_i'}$. Similarly, one finds that $|G/p^iG| = p^{\mu_1' + \cdots + \mu_i'}$ from the lemma. Hence,

$$|G/G_i| \leq |G/p^iG| \Rightarrow \lambda_1' + \cdots + \lambda_i' \leq \mu_1' + \cdots + \mu_i'$$

Thus, $\lambda' \leq \mu'$ implies $\mu \leq \lambda$.

Now, if $\mu = \lambda$, then the only possible chain is

$$G \supseteq pG \supseteq p^2G \supseteq \cdots \supseteq \{0\},$$

because of the condition $p^iG \subseteq G_i$. Thus, we find that $a_{\lambda\mu} = 0$, unless $\mu \leq \lambda$ and $a_{\lambda\lambda} = 1$.

Therefore, the matrix $(a_{\lambda\mu})$ is strictly upper unitriangular, whence we can solve for the u_μ as integral linear combinations of the u_{π_λ}'s.

Moreover, the u_{π_λ}'s are $\mathbb{Z}$-linear independent, whence the elements $u_{(1^r)}$ are algebraically independent over $\mathbb{Z}$.

$\square$

As a result, the Hall algebra H and the ring Λ of symmetric functions are isomorphic.

By viewing both H and Λ as $\mathbb{Q}$-algebras, an interesting choice of isomorphism is the following $\mathbb{Q}$-linear map:

$$\begin{aligned} \psi : H &\longrightarrow \Lambda \\ u_{(1^r)} &\mapsto p^{-\frac{r(r-1)}{2}} \cdot e_r(x) . \end{aligned}$$

We shall see that the Hall-Littlewood polynomial $P_\lambda(x, p^{-1})$ appears in the image of u_λ through ψ.